FUNCTORIALITY OF THE COMPLETION FUNCTOR UNDER BASE CHANGE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file restricted.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the counted statement is the author lemma environment labelled lemma-functoriality-completion-functor; the proof environment immediately following it is its own proof; the context span delivers the section situation and the completion functor equation that 'in the situation above' refers to

Editorial note. The label lemma-functoriality-completion-functor names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- the section-algebraization setup: locally Noetherian X, closed T, U, and the completion functor (h : Y to X) mapping to Y_{/T}, together with the category definitions (delivered as context)
- the completion functor commutes with fibre products (formal-spaces-section-completion-functor)

Source reference. Original reference 0AR2: sources/stacks-restricted/pinned-original-source.tex, source lines 6186–6186. Prerequisite pointer invoked at 6878; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 0GDB: sources/stacks-restricted/pinned-original-source.tex, source lines 5953–5953. Prerequisite pointer invoked at 6832; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

The notation in this section will agree with the notation in Section 0GDB except our algebraic spaces and formal algebraic spaces will be locally Noetherian.

Thus, we first fix a base scheme S. All rings, topological rings, schemes, algebraic spaces, and formal algebraic spaces and morphisms between these will be over S. Next, we fix a locally Noetherian algebraic space X and a closed subset $T \subset |X|$. We denote $U \subset X$ be the open subspace with $|U| = |X| \setminus T$. Picture

$$U \to X \quad |X| = |U| \amalg T$$

Given a morphism of algebraic spaces $f : X' \to X$, we will use the notation $U' = f^{-1}U$, $T' = |f|^{-1}(T)$, and $f_{/T} : X'_{/T'} \to X_{/T}$ as in Section 0GDB. We will sometimes write $X'_{/T}$ in stead of $X'_{/T'}$ and more generally for a morphism a :

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$X' \to X''$ of algebraic spaces over X we will denote $a_{/T} : X'_{/T} \to X''_{/T}$ the induced morphism of formal algebraic spaces obtained by completing the morphism a along the inverse images of T in X' and X''.

Given this setup we will consider the functor

$$\left\{\begin{array}{c}\text{morphisms of algebraic spaces} \\ f : X' \to X \text{ which are locally} \\ \text{of finite type and such that} \\ U' \to U \text{ is an isomorphism}\end{array}\right\} \longrightarrow \left\{\begin{array}{c}\text{morphisms } g : W \to X_{/T} \\ \text{of formal algebraic spaces} \\ \text{with } W \text{ locally Noetherian} \\ \text{and } g \text{ rig-étale}\end{array}\right\} \tag{0.0.1}$$

sending $f : X' \to X$ to $f_{/T} : X'_{/T'} \to X_{/T}$. This makes sense because $f_{/T}$ is rig-étale by Lemma 0AR2.

Lemma 0.1. *In the situation above, let $X_1 \to X$ be a morphism of algebraic spaces with X_1 locally Noetherian. Denote $T_1 \subset |X_1|$ the inverse image of T and $U_1 \subset X_1$ the inverse image of U. We denote*

1. *$\mathcal{C}_{X,T}$ the category whose objects are morphisms of algebraic spaces $f : X' \to X$ which are locally of finite type and such that $U' = f^{-1}U \to U$ is an isomorphism,*
2. *$\mathcal{C}_{X_1,T_1}$ the category whose objects are morphisms of algebraic spaces $f_1 : X'_1 \to X_1$ which are locally of finite type and such that $f_1^{-1}U_1 \to U_1$ is an isomorphism,*
3. *$\mathcal{C}_{X_{/T}}$ the category whose objects are morphisms $g : W \to X_{/T}$ of formal algebraic spaces with W locally Noetherian and g rig-étale,*
4. *$\mathcal{C}_{X_{1,/T_1}}$ the category whose objects are morphisms $g_1 : W_1 \to X_{1,/T_1}$ of formal algebraic spaces with W_1 locally Noetherian and g_1 rig-étale.*

Then the diagram

$$\begin{array}{ccc} \mathcal{C}_{X,T} & \longrightarrow & \mathcal{C}_{X_{/T}} \\ \downarrow & & \downarrow \\ \mathcal{C}_{X_1,T_1} & \longrightarrow & \mathcal{C}_{X_{1,/T_1}} \end{array}$$

is commutative where the horizontal arrows are given by (0.0.1) and the vertical arrows by base change along $X_1 \to X$ and along $X_{1,/T_1} \to X_{/T}$.

Proof. This follows immediately from the fact that the completion functor $(h : Y \to X) \mapsto Y_{/T} = Y_{/|h|^{-1}T}$ on the category of algebraic spaces over X commutes with fibre products. □